David Lydston

Principal Product Manager | Data Platforms, Metric Governance & Experimentation

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Profile

Principal Product Manager with 10+ years building experimentation, measurement, and big-data analytics platforms at Microsoft. Leads a shared scorecard platform serving **30+ products** and more than **3,000 experiments each month**, with division-wide evaluation using common metrics drawn exclusively from the UDM measurement catalog. Promotes catalog adoption and championed adding experiment randomizations for **30+ products**. Standardized measurement across hundreds of features and delivered **$7M in scorecard cost improvements** (May–Sep 2024). Skilled at managing multi-discipline teams and aligning senior partners across Bing, Office, and Azure.

Core expertise

Causal data analysis · Metric governance & standardization · Experimentation platforms & scorecards · Data-quality monitoring · Big-data compute efficiency (Cosmos, Spark) · Azure compliance & production readiness · Sensitive-data protection · Product strategy & roadmaps · Executive alignment

Professional experience

Principal PM, IDEAS Experimentation Lead

Microsoft | Sep 2024 – Present

Lead division-wide standardization of experiment evaluation across products and platforms using a common set of metrics. Require all scorecard metrics to come from the UDM measurement catalog, promoting its adoption and contributing to its evolution.

- Built a cross-organization experiment scorecard platform serving **30+ products** and more than **3,000 experiments each month**.
- Championed adding experiment randomizations to the UDM catalog for **30+ products**.
- Led migration of key metrics from legacy to team-owned scorecards, standardizing measurement across **hundreds of features** supporting major launches.
- Restored Copilot experimentation scorecards across about 14 Microsoft 365 partner areas, establishing a single source of truth for ownership, readiness, and service targets.
- Reduced the experimentation SLA from **EOD + 3 days to EOD + 36 hours** and cut scorecard compute consumption **more than 75%** through fast metrics and lean scorecards.
- Grew scorecard coverage about **50x**, from one experiment at a time to more than **85% of E+D experimentation scale**.
- Relaunched the Expedite homepage as a leadership decision system, growing experiment coverage **50%** and surfacing 10,238 active experiments.
- Rejected requests to place High Business Impact (HBI) data on unsecured scorecards, protecting security posture during platform transitions.

Principal PM Manager, Office Growth Accelerators Experimentation

Microsoft | Oct 2023 – Sep 2024

- Prioritized a Copilot token-usage scorecard across all Office experiments and rollouts, catching a **7% top-of-funnel regression** in OneNote ($p < 0.001$) that would otherwise have been missed.
- Reduced time to insight **20%** for Office Online experiments and enabled Word Online scorecards **14 hours earlier**.
- Co-authored the FY25 experimentation strategy, recognized by senior leadership as the organization's vision.

Principal PM, Experimentation Lead, Office Growth Accelerators

Microsoft | Nov 2022 – Oct 2023

- Restored platform trust with Microsoft's lowest sample-ratio-mismatch rate, a scorecard alert-to-bug pathway, and partner office hours.
- Resolved a recurring escalation by showing with data that scorecard delivery was not a meaningful driver of experiment velocity.

Principal Program Manager, Analysis & Experimentation

Microsoft | Jan 2020 – Nov 2022 | Promoted to Principal PM Jan 2020

- Prioritized and planned the experimentation platform's five core Azure certifications and Ring 1 readiness, unblocking Dynamics 365 onboarding and Azure Portal adoption.
- Negotiated an Azure Compliance audit exception that removed a policy blocker to platform certification.
- Co-drove a **100,000 daily Cosmos compute-token reduction** with Bing, cutting ship-decision time **from six days to two**.
- Negotiated cost sharing with Bing to fund a Cosmos-to-Spark migration and staged delivery when proposed scope grew 50%.
- Authored and externally published a cross-product experimentation cookbook adopted by multiple Microsoft organizations.

Senior PM, Analysis & Experimentation

Microsoft | May 2018 – Jan 2020

- Improved pipeline fault tolerance **10x** and cut p90 job-submission latency from over an hour to about 15 minutes.
- Improved intraday alert accuracy **50%** with bot-tolerant, user-based metrics, virtually eliminating false positives.
- Shipped metric vectorization, delivering about **30% Cosmos compute-token efficiency** for trigger and filter scorecards.
- Delivered **33% Exchange scorecard efficiency** by eliminating an approximately 4x compute-token over-allocation.

Senior PM, Microsoft Marketing Analytics

Microsoft | Feb 2016 – May 2018

- Brought Windows 10 DAU/WAU/MAU by platform into SDX, correcting a frequently cited erroneous insights source.
- Reduced Adobe Analytics sampling volume **50%**, delivering **$60K+/month in savings** while preserving business-review cadence.
- Hardened analytics access by retiring 28 administrator accounts, ending account sharing, and enforcing global address library validation.

Leadership

Set vision and charter for a 24-person experimentation team (15 engineers, five data scientists, and four product managers) through organizational changes and a team reduction. Directly managed four individual contributors.

Education

Master of Science, Information Systems Management
Seattle Pacific University | 2005–2007

Master of Arts, Behavioral Psychology
San Diego State University | 1997–1999

Bachelor of Arts, Psychology
Western Washington University | 1994–1997